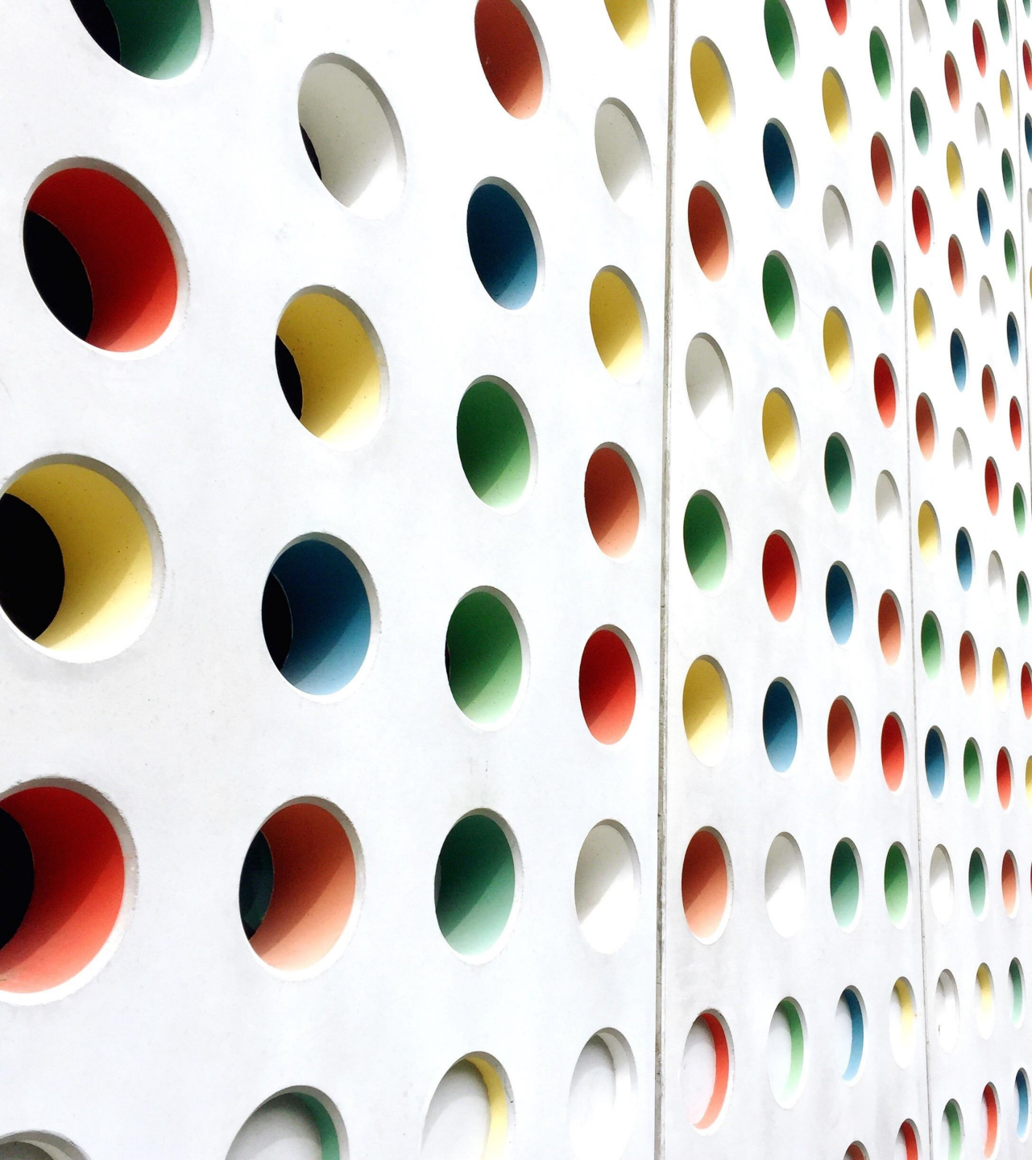

SPARQL

BY

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Conventions

Red text means:

“This is a core part of the SPARQL syntax or language.”

Blue text means:

“This is an example of query-specific text or values that might go into a SPARQL query.”

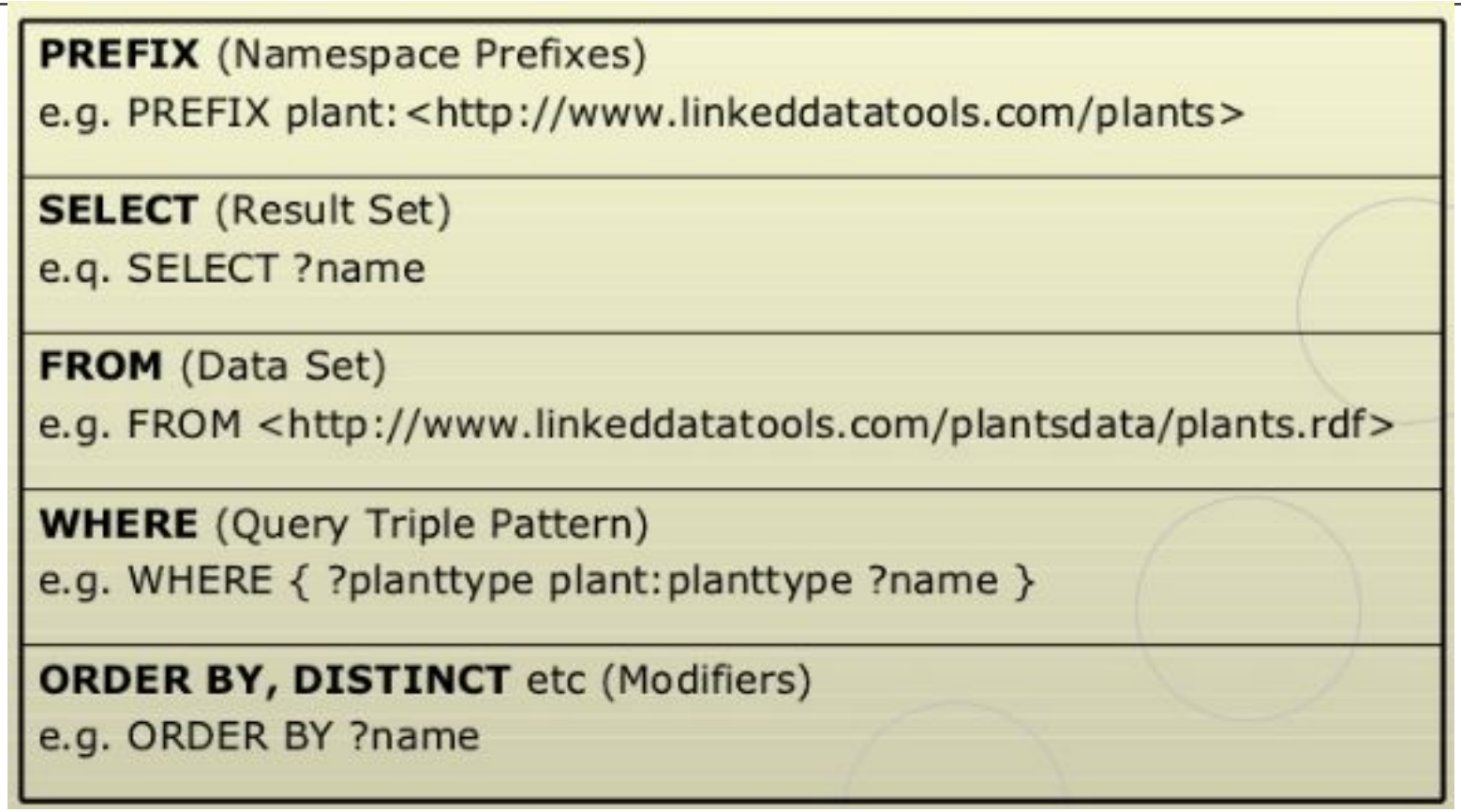
- SPARQL is a query language/engine for RDF graphs.
- An RDF graph is a set of triples. A flexible and extensible way to represent information about WWW resources
- A concept similar to SQL for data bases. A W3C standard query language to fetch data from distributed Semantic Web data models.
- Can query a triple store or data on the Web (at a given URL). It provides facilities to:
  - extract information in the form of URIs, blank nodes, plain and typed literals.
  - extract RDF subgraphs.
  - construct new RDF graphs based on information in the queried graphs

Sparql – FIRST EXAMPLE

```
prefix sch-ont: <http://education.data.gov.uk/def/school/>
SELECT ?name WHERE { ?school a sch-ont:School;
      sch-ont:establishmentName ?name;
      sch-ont:districtAdministrative <http://statistics.data.gov.uk/id/local-authority-district/00AA> ;
      }
ORDER BY ?name
```

- If executed on the UK government's Open Semantic Database, will return the names of all the schools in the UK in administrative district 00AA, and order the results in alphabetical order (http://data.gov.uk/sparql)
- **SPARQL Is Similar To SQL**
  - Like SQL, SPARQL selects data from the query data set by using a SELECT statement to determine which subset of the selected data is returned. Also, SPARQL uses a WHERE clause to define graph patterns to find a match for in the query data set.

General clause



Data set – triple [subject, predicate, object]

Subject	Predicate	Object
http://www.linkeddatatools.com /plants#magnolia	http://www.linkeddatatools.com /plants#family	Magnoliaceae
http://www.linkeddatatools.com /plants#african_lilly	http://www.linkeddatatools.com /plants#family	Liliaceae
http://www.linkeddatatools.com /plants#silvertop	http://www.linkeddatatools.com /plants#family	Aralianae
http://www.linkeddatatools.com /plants#velvetleaf	http://www.linkeddatatools.com /plants#family	Malvaceae
http://www.linkeddatatools.com /plants#manglietia	http://www.linkeddatatools.com /plants#family	Magnoliaceae

Example: triple data containing a variety of shrubs and plants, and their family names

Select All Data

```
PREFIX plants: <http://www.linkeddatatools.com/plants>
SELECT * WHERE
{ ?name plants:family ?family }
```

- select all the plant URIs (subjects) and plant family names (literal-type objects) from the data above;
- the wildcard '*' in SPARQL, similar to SQL, will return all the mapped data in the result set;
- as we have stated two variables ?name and ?family, the query will return both these declared query variables in our result set, according to the subjects, predicates or objects they're mapped to;

Nuts & Bolts

URIs

Write full URIs:

```
<http://this.is.a/full/URI/written#out>
```

Abbreviate URIs with prefixes:

```
PREFIX foo: <http://this.is.a/URI/prefix#>
… foo:bar …
      ➪  http://this.is.a/URI/prefix#bar
```

Shortcuts:

```
a  ➪  rdf:type
```

Literals

Plain literals:

```
"a plain literal"
```

Plain literal with language tag:

```
"bonjour"@fr
```

Typed literal:

```
"13"^^xsd:integer
```

Shortcuts:

```
true ➪ "true"^^xsd:boolean
3    ➪ "3"^^xsd:integer
4.2  ➪ "4.2"^^xsd:decimal
```

Variables

Variables:

```
?var1, ?anotherVar, ?and_one_more
```

Comments

Comments:

```
# Comments start with a '#'
# continue to the end of the line
```

Triple Patterns

Match an exact RDF triple:

```
ex:myWidget ex:partNumber "XY24Z1" .
```

Match one variable:

```
?person foaf:name "Lee Feigenbaum" .
```

Match multiple variables:

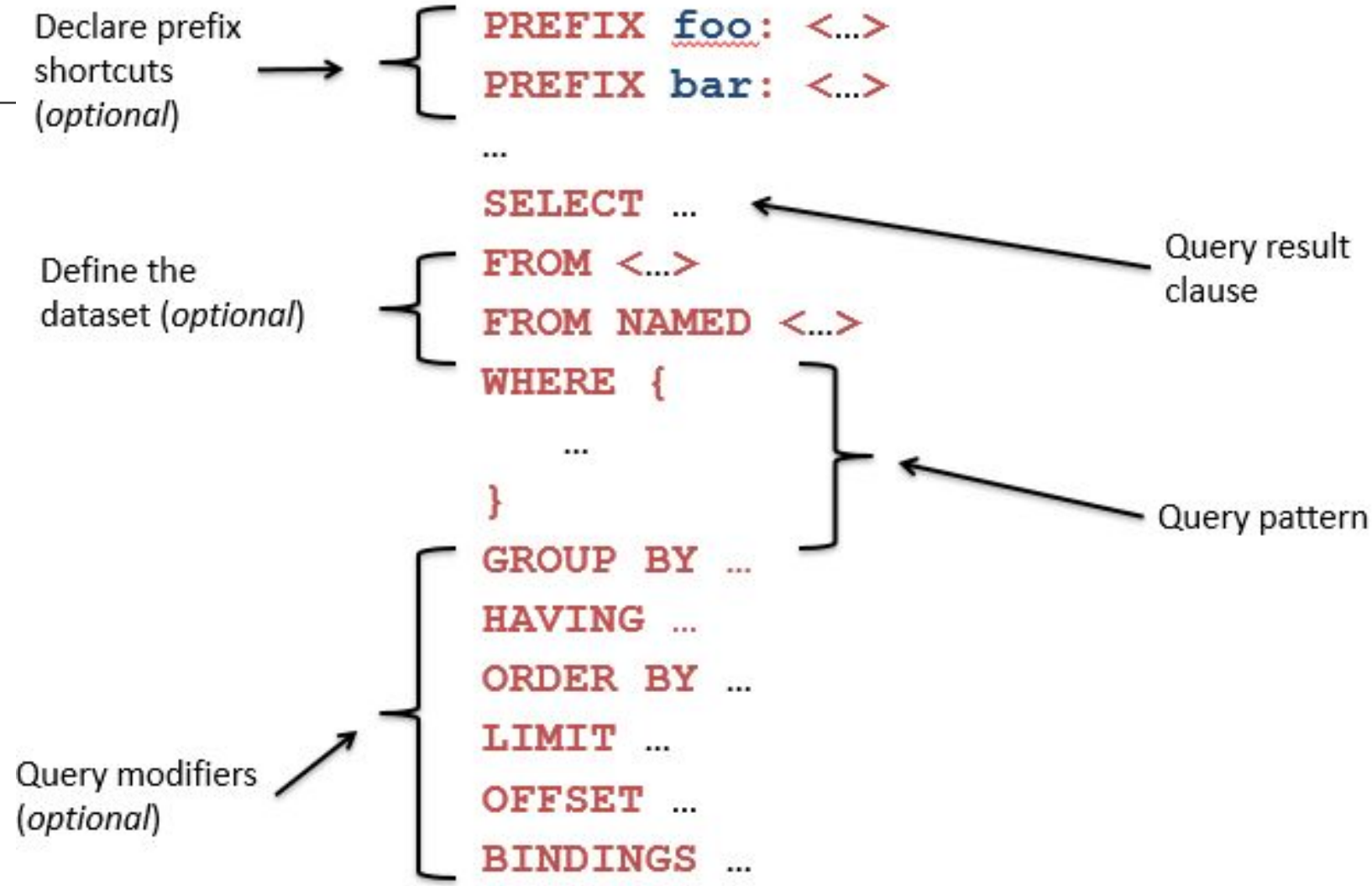
```
conf:SemTech2009 ?property ?value .
```

Common Prefixes

prefix...	…stands for
rdf:	http://xmlns.com/foaf/0.1/
rdfs:	http://www.w3.org/2000/01/rdf-schema#
owl:	http://www.w3.org/2002/07/owl#
xsd:	http://www.w3.org/2001/XMLSchema#
dc:	http://purl.org/dc/elements/1.1/
foaf:	http://xmlns.com/foaf/0.1/

More common prefixes at http://prefix.cc

Anatomy of a Query



4 Types of SPARQL Queries

SELECT queries

Project out specific variables and expressions:

```
SELECT ?c ?cap (1000 * ?people AS ?pop)
```

Project out all variables:

```
SELECT *
```

Project out distinct combinations only:

```
SELECT DISTINCT ?country
```

Results in a table of values (in XML or JSON):

?c	?cap	?pop
ex:France	ex:Paris	63,500,000
ex:Canada	ex:Ottawa	32,900,000
ex:Italy	ex:Rome	58,900,000

CONSTRUCT queries

Construct RDF triples/graphs:

```
CONSTRUCT {
   ?country a ex:HolidayDestination ;
       ex:arrive_at ?capital ;
       ex:population ?population .
}
```

Results in RDF triples (in any RDF serialization):

```
ex:France a ex:HolidayDestination ;
    ex:arrive_at ex:Paris ;
    ex:population 635000000 .
ex:Canada a ex:HolidayDestination ;
    ex:arrive_at ex:Ottawa ;
    ex:population 329000000 .
```

ASK queries

Ask whether or not there are any matches:

```
ASK
```

Result is either "true" or "false" (in XML or JSON):

```
true, false
```

DESCRIBE queries

Describe the resources matched by the given variables:

```
DESCRIBE ?country
```

Result is RDF triples (in any RDF serialization) :

```
ex:France a geo:Country ;
  ex:continent geo:Europe ;
  ex:flag <http://…/flag-france.png> ;
  …
```

Combining SPARQL Graph Patterns

*Consider **A** and **B** as graph patterns.*

A Basic Graph Pattern – one or more triple patterns

```
A . B
```

➪ Conjunction. Join together the results of solving A and B by matching the values of any variables in common.

Optional Graph Patterns

```
A OPTIONAL { B }
```

➪ Left join. Join together the results of solving A and B by matching the values of any variables in common, if possible. Keep all solutions from A whether or not there's a matching solution in B

Combining SPARQL Graph Patterns

*Consider **A** and **B** as graph patterns.*

Either-or Graph Patterns

```
{ A } UNION { B }
```

➪ Disjunction. Include both the results of solving A and the results of solving B.

“Subtracted” Graph Patterns (SPARQL 1.1)

```
A MINUS { B }
```

➪ Negation. Solve A. Solve B. Include only those results from solving A that are *not compatible* with any of the results from B.

SPARQL Subqueries *[SPARQL 1.1]*

*Consider **A** and **B** as graph patterns.*

```
A .
{
   SELECT …
   WHERE {
      B
   }
}
C .
```

- Join the results of the subquery with the results of solving A and C.

SPARQL Filters

SPARQL **FILTER***s eliminate solutions that do not cause an expression to evaluate to true.*

```
A . B . FILTER ( ...expr... )
```

Place **FILTER***s in a query inline within a basic graph pattern*

Category	Functions / Operators	Examples
Logical	`!, &&, ||, =, !=, <, <=, >, >=`	`?hasPermit || ?age < 25`
Math	`+, -, *, /`	`?decimal * 10 > ?minPercent`
Existence *(SPARQL 1.1)*	`EXISTS, NOT EXISTS`	`NOT EXISTS { ?p foaf:mbox ?email }`
SPARQL tests	`isURI, isBlank, isLiteral, bound`	`isURI(?person) || !bound(?person)`
Accessors	`str, lang, datatype`	`lang(?title) = "en"`
Miscellaneous	`sameTerm,`	`regex(?ssn, "\\d{3}-\\d{2}-\\d{4}")`

Aggregates *[SPARQL 1.1]*

?key	?val	?other1
1	4	…
1	4	…
2	5	…
2	4	…
2	10	…
2	2	…
2	1	…
3	3	…

1. Partition results into groups based on the expression(s) in the **GROUP BY** clause

?key	?sum_of_val
1	8
2	22
3	3

2. Evaluate projections and aggregate functions in **SELECT** clause to get one result per group

?key	?sum_of_val
1	8
3	3

3. Filter aggregated results via the **HAVING** clause

SPARQL 1.1 includes: **COUNT**, **SUM**, **AVG**, **MIN**, **MAX**, **SAMPLE**, **GROUP_CONCAT**

Property Paths *[SPARQL 1.1]*

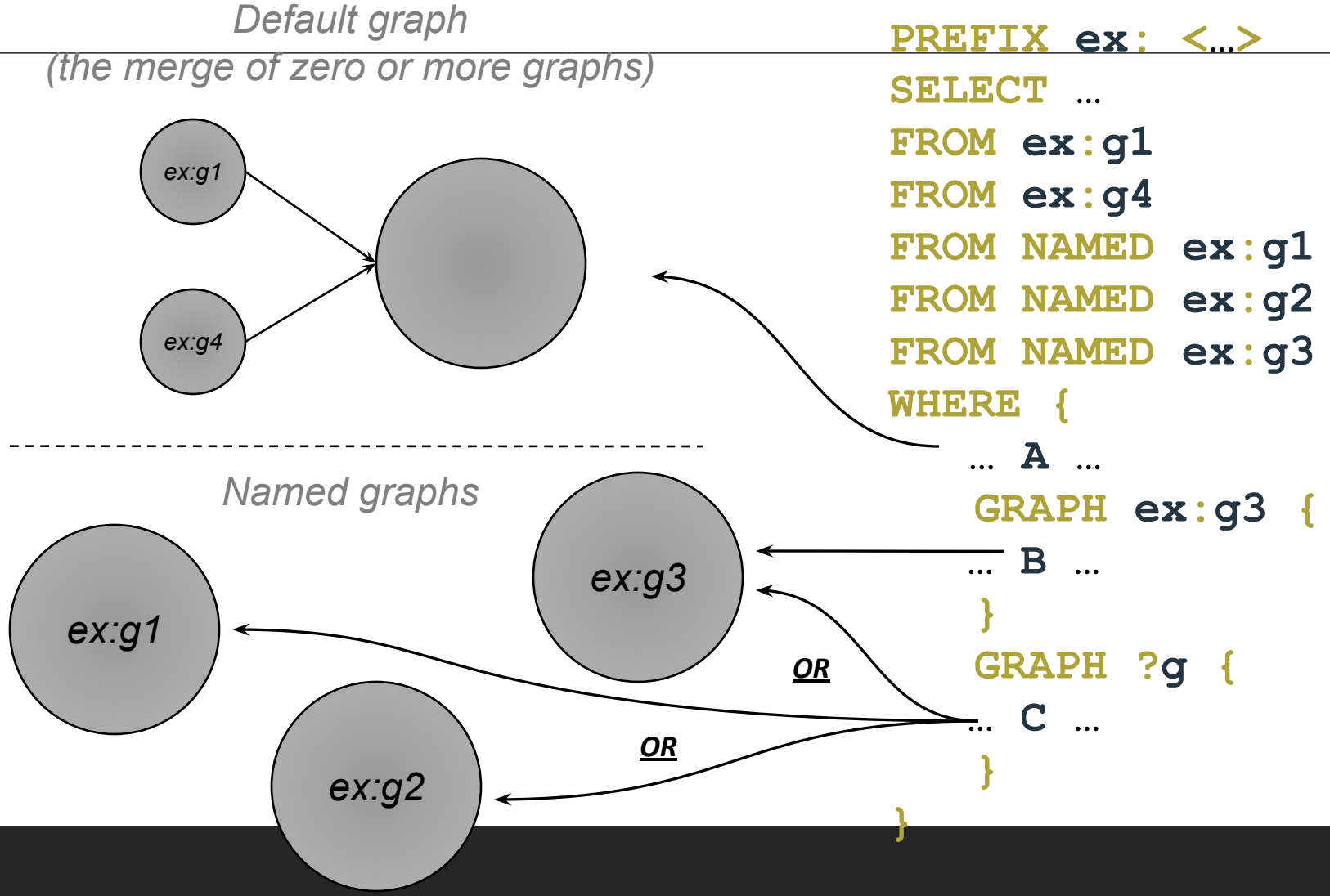
Property paths allow triple patterns to match arbitrary-length paths through a graph

Predicates are combined with regular-expression-like operators:

Construct	Meaning
`path1/path2`	Forwards path (`path1` followed by `path2`)
`^path1`	Backwards path (object to subject)
`path1|path2`	Either `path1` or `path2`
`path1*`	`path1`, repeated zero or more times
`path1+`	`path1`, repeated one or more times
`path1?`	`path1`, optionally
`path1{m,n}`	At least `m` and no more than `n` occurrences of `path1`
`path1{n}`	Exactly `n` occurrences of `path1`
`path1{m,}`	At least `m` occurrences of `path1`
`path1{,n}`	At most `n` occurrences of `path1`

RDF Datasets

A SPARQL queries a *default graph* (normally) and zero or more *named graphs* (when inside a **GRAPH** clause).




```
PREFIX ex: <…>
SELECT …
FROM ex:g1
FROM ex:g4
FROM NAMED ex:g1
FROM NAMED ex:g2
FROM NAMED ex:g3
WHERE {
   … A …
   GRAPH ex:g3 {
   … B …
   }
   GRAPH ?g {
   … C …
   }
}
```

SPARQL in 11 mins

https://www.youtube.com/watch?v=FvGndkpa4K0

SPARQL Resources

- The SPARQL Specification
  - http://www.w3.org/TR/rdf-sparql-query/
- SPARQL implementations
  - http://esw.w3.org/topic/SparqlImplementations
- SPARQL endpoints
  - http://esw.w3.org/topic/SparqlEndpoints
- SPARQL Frequently Asked Questions
  - http://www.thefigtrees.net/lee/sw/sparql-faq
- SPARQL Working Group
  - http://www.w3.org/2009/sparql/wiki/
- Common SPARQL extensions
  - http://esw.w3.org/topic/SPARQL/Extensions